

Renewable Energy Sector: A Comprehensive 2024 Review

This report covers three distinct dimensions of the renewable energy sector's 2024 performance: generation capacity growth (solar and wind), enabling storage technology cost trends, and workforce expansion across the sector's major job categories. Each dimension is analyzed separately, as growth in one does not necessarily track growth in the others.

1. Generation Capacity

1.1 Overview

Renewable generation capacity has grown substantially faster than forecast in most major markets since 2022, driven by declining solar panel manufacturing costs and a wave of national subsidy programs following the 2022 global energy price shock. Wind capacity grew more steadily, constrained less by cost and more by offshore permitting timelines.

The chart below shows the divergence between solar and wind capacity growth over the six-year study window. Solar, which trailed wind for most of the period, overtook it in 2024 for the first time.

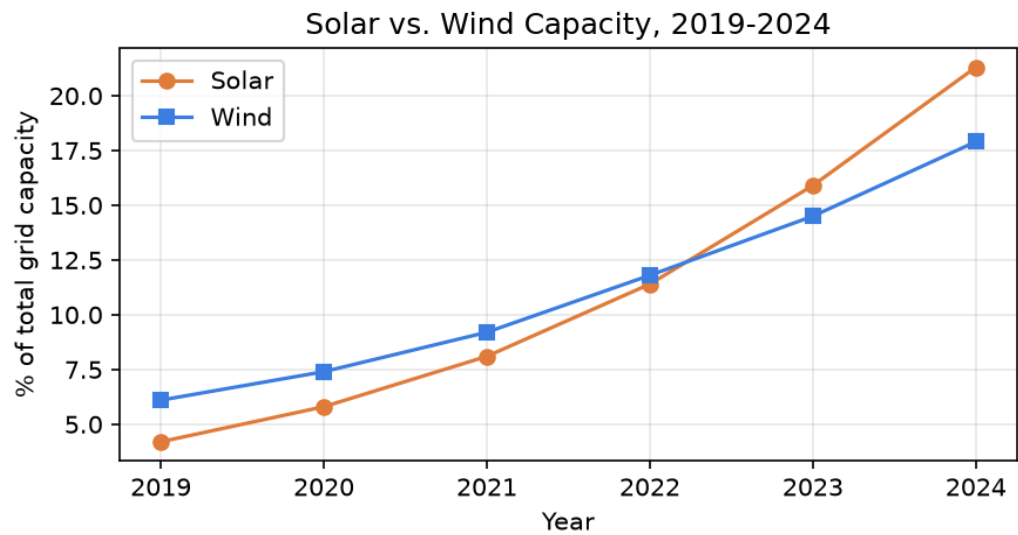


Figure 1: Global installed solar and wind capacity as a percentage of total grid capacity, 2019-2024.

1.2 Regional Adoption Rates

Adoption rates vary significantly by region. The table below reports each region's renewable share of total electricity generation for 2023 and 2024.

Region	2023 Share	2024 Share	YoY Change
Northern Europe	58.2%	64.7%	+6.5 pp
East Asia	31.4%	39.1%	+7.7 pp
North America	27.8%	32.0%	+4.2 pp
South Asia	19.6%	24.3%	+4.7 pp
Sub-Saharan Africa	14.1%	16.8%	+2.7 pp
Oceania	35.0%	41.5%	+6.5 pp

East Asia recorded the largest year-over-year gain at 7.7 percentage points, driven primarily by accelerated solar deployment.

2. Storage Technology

Grid-scale battery storage cost has fallen sharply over the study period, a critical enabler for renewable adoption since storage capacity determines how much variable solar and wind generation a grid can absorb without curtailment.

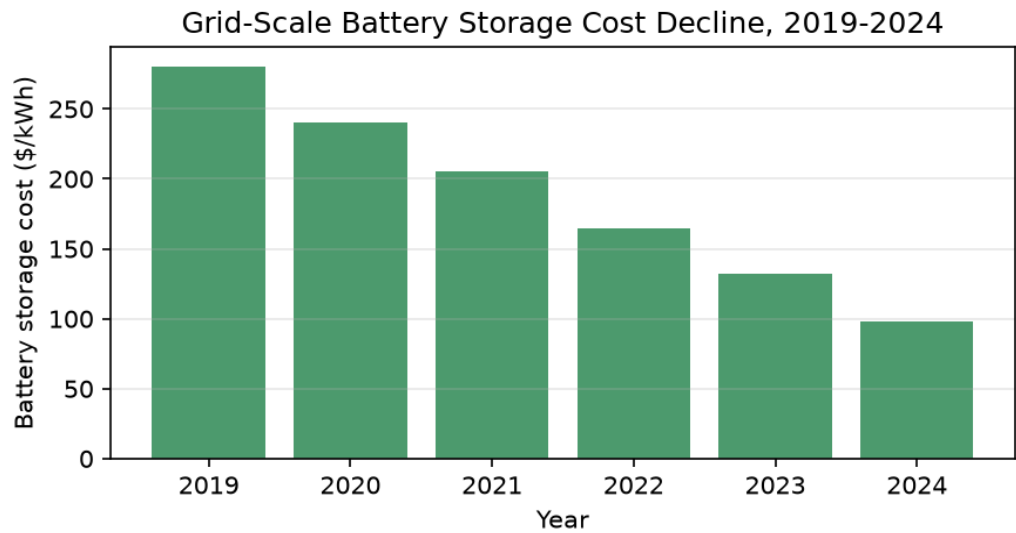


Figure 2: Grid-scale lithium-ion battery storage cost per kWh, 2019-2024.

Storage cost declined from \$280/kWh in 2019 to \$98/kWh in 2024, a 65% reduction. Analysts attribute roughly half of this decline to manufacturing scale and half to chemistry improvements in cathode materials.

2.1 Storage Deployment by Technology Type

The table below breaks down 2024 grid-scale storage deployment by technology, measured in gigawatt-hours (GWh) of installed capacity.

Technology	2023 GWh Installed	2024 GWh Installed	Growth
Lithium-ion	142	218	+53.5%
Flow battery	8	19	+137.5%
Pumped hydro	310	324	+4.5%
Compressed air	3	7	+133.3%

Flow battery and compressed-air storage grew fastest in percentage terms, though both remain small in absolute capacity compared to pumped hydro and lithium-ion.

3. Workforce and Investment

Sector employment grew across all major job categories in 2024, with installation roles adding the most positions in absolute terms.

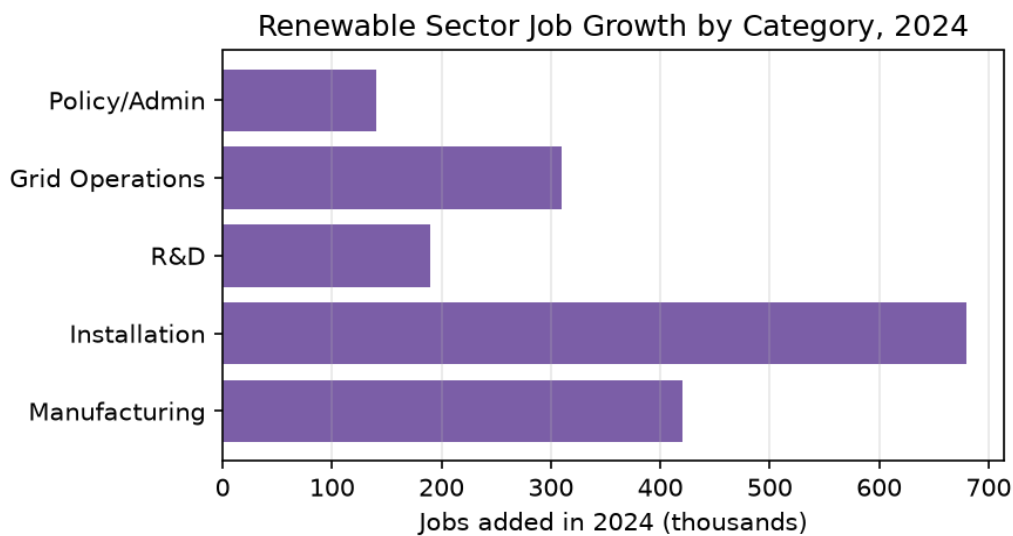


Figure 3: New renewable-sector jobs added by category, 2024, in thousands.

Installation added 680,000 jobs in 2024, more than any other category, reflecting the labor-intensive nature of solar and wind construction relative to manufacturing, which benefits more from automation.

3.1 Regional Investment Totals

Total capital investment by region in 2024, in billions of US dollars.

Region	2023 Investment (\$B)	2024 Investment (\$B)	Growth
Northern Europe	84	112	+33.3%
East Asia	156	231	+48.1%
North America	98	127	+29.6%
South Asia	41	58	+41.5%

East Asia recorded both the highest absolute investment and the highest percentage growth in 2024, consistent with its lead in regional capacity adoption growth described in Section 1.2.

4. Outlook

Analysts project continued acceleration through 2026 across all three dimensions covered in this report. Solar capacity growth is expected to keep outpacing wind due to shorter installation timelines. Storage costs are projected to fall below \$80/kWh by 2026 as flow battery manufacturing reaches commercial scale. Workforce growth is expected to shift increasingly toward grid operations roles as installed base maintenance needs grow relative to new construction.